

MS 6015: Multivariate Statistical Methods

Regression Analysis — Simple, Multiple, and Advanced Topics

Condensed Lecture Notes — Reference: Stine & Foster (Statistics for Business);
Keller (Managerial Statistics)

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1 Simple Linear Regression

1.1 Introduction and Motivation

Regression analysis is the most widely used statistical tool for understanding relationships among variables. It connects a **response (dependent) variable** Y to one or more **explanatory (predictor) variables** X via an equation.

Typical business questions:

- What is the impact of an MBA on income?
- How do mutual fund returns relate to the market? (CAPM)
- Does a company discriminate against women in salary?
- Predict house price from observed characteristics.

1.2 Describing Association

Before fitting a model, examine a scatterplot and ask four questions:

1. **Direction** — does the pattern trend up or down?
2. **Curvature** — is the pattern linear or curved?
3. **Variation** — are points tightly or loosely clustered?
4. **Outliers** — is there anything unexpected?

Association is measured numerically by **covariance** and **correlation**.

1.3 The Simple Regression Model (SRM)

The SRM describes how the conditional mean of Y depends on X :

$$Y = \beta_0 + \beta_1 X + \epsilon, \quad \epsilon \sim N(0, \sigma_\epsilon^2).$$

Three Assumptions about ϵ

1. **Independence.** Errors are independent of each other.
2. **Equal variance (homoscedasticity).** $\text{Var}(\epsilon) = \sigma_\epsilon^2$ (constant).
3. **Normality.** Errors are normally distributed.

1.4 Least Squares Estimation

The **least squares** estimates minimise $\sum_{i=1}^n e_i^2$, yielding:

$$b_1 = r_{XY} \frac{s_Y}{s_X}, \quad b_0 = \bar{Y} - b_1 \bar{X}.$$

where r_{XY} is the sample correlation and s_X, s_Y are standard deviations.

The residual is the difference between observed and fitted value:

$$e_i = Y_i - \hat{Y}_i, \quad \hat{Y}_i = b_0 + b_1 X_i.$$

Key properties: $\text{corr}(X, e) = 0$ and $\text{corr}(\hat{Y}, e) = 0$.

1.5 Variance Decomposition and R^2

$$\underbrace{\sum (Y_i - \bar{Y})^2}_{\text{SST}} = \underbrace{\sum (\hat{Y}_i - \bar{Y})^2}_{\text{SSR (Regression)}} + \underbrace{\sum e_i^2}_{\text{SSE (Error)}}.$$

The **coefficient of determination**:

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}} = r_{XY}^2.$$

$R^2 \in [0, 1]$; closer to 1 means better fit. If $R^2 = 1$, perfect fit.

1.6 Inference in Simple Regression

The standard error of the slope:

$$\text{se}(b_1) = \frac{s_e}{s_X \sqrt{n-1}}, \quad \text{where } s_e = \sqrt{\frac{\text{SSE}}{n-2}}.$$

$\text{se}(b_1)$ increases with residual spread, decreases with sample size and spread of X .

95% confidence interval for β_1 : $b_1 \pm t_{n-2, 0.025} \cdot \text{se}(b_1)$.

Hypothesis test $H_0 : \beta_j = 0$: use $t = b_j / \text{se}(b_j)$. Reject H_0 if $|t| > 2$ (approx.) or p-value < 0.05 .

Three Equivalent Ways to Reject $H_0 : \beta_j = 0$

- Zero lies outside the 95% confidence interval.
- $|t\text{-statistic}| > 2$.
- p-value < 0.05 .

1.7 Capital Asset Pricing Model (CAPM) Example

The CAPM is formulated as a simple regression:

$$R_t = \beta_0 + \beta_1 M_t + \epsilon_t,$$

where R_t is the asset return and M_t is the market return. Theory predicts $\beta_0 = 0$.

For Berkshire Hathaway: slope = 0.722, $t = 9.29$, $p < 0.0001$ (slope significant); intercept = 1.396, $t = 4.11$, $p < 0.0001$ (intercept also significant, inconsistent with pure CAPM).

1.8 Prediction Intervals

A **95% prediction interval** for a new observation y_{new} at x_{new} is:

$$\hat{y}_{\text{new}} \pm t_{n-2, 0.025} \cdot \text{se}(\hat{y}_{\text{new}}), \quad \text{se}(\hat{y}_{\text{new}}) = s_e \sqrt{1 + \frac{1}{n} + \frac{(x_{\text{new}} - \bar{x})^2}{(n-1)s_X^2}}.$$

Simple approximation: $\hat{y} \pm 2s_e$.

Prediction Interval vs. Confidence Interval

A *confidence interval* estimates the mean response $E(Y|X = x)$ — a population parameter. A *prediction interval* bounds a *single new observation* — it is always wider because it accounts for individual variability as well.

2 Curved Patterns and Transformations

2.1 Detecting Nonlinearity

- Plot Y vs. X : look for curvature.
- Plot residuals vs. X : a systematic curve in residuals confirms nonlinearity.
- Ask: do equal-sized changes in X produce equal changes in Y ? If not, the relationship is nonlinear.

2.2 Transformations

A **transformation** re-expresses a variable by applying a function (e.g. log, reciprocal) to linearise the relationship.

- **Reciprocal:** convert $d \rightarrow 1/d$. Useful for ratio variables (e.g. miles per gallon $\rightarrow$ gallons per 100 miles).
- **Logarithm:** useful for exponential growth or right-skewed data.

Choosing a transformation: match the curvature pattern in the scatterplot to a standard shape, then pick the transformation that straightens it.

Key rule: only compare R^2 values between models that use the *same response variable*.

3 Regression Diagnostics

Three major problems to diagnose: changing variation, leveraged outliers, and dependent errors.

Assumption	Diagnostic	Problem if Violated
Linearity	Scatterplot	Nonlinear pattern
Independence	Residuals vs. time	Autocorrelation
Constant Variance	Residual plot	Heteroscedasticity
Normality	Normal quantile plot	Outliers, skewness

3.1 Changing Variation (Heteroscedasticity)

Detection: fan-shaped residual plot; side-by-side boxplots showing increasing spread.

Consequences:

- Prediction intervals are too narrow for large X and too wide for small X .
- Confidence intervals and hypothesis tests for β_0, β_1 are unreliable.

Fix: transform the response (e.g. regress price per square foot on 1/SqFt to stabilise variance in house price models).

3.2 Leveraged Outliers

A **leveraged observation** has an unusually small or large value of X . It can pull the regression line toward itself, distorting slope and intercept estimates and making prediction intervals unreliable.

Approach: fit the model with and without the outlier; compare estimates using standard errors as the scale. Decide whether to include based on whether the outlier is representative of future data.

3.3 Dependent Errors and Time Series

When data are collected over time, residuals may be correlated (**autocorrelation**).

Detection: plot residuals vs. time; use the **Durbin-Watson statistic**:

$$D = \frac{\sum_{t=2}^n (e_t - e_{t-1})^2}{\sum_{t=1}^n e_t^2}.$$

Roughly: $D < 1.5$ suggests positive autocorrelation; $D > 2.5$ suggests negative autocorrelation.

Consequences: standard errors are underestimated, making tests too liberal.

Fix: incorporate the dependence structure into the model.

4 Multiple Linear Regression (MLR)

4.1 The Model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon, \quad \epsilon \sim N(0, \sigma^2).$$

The conditional mean of Y is $E(Y|X_1, \dots, X_p) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$.

4.2 Least Squares Estimates

Minimise $\sum e_i^2$ over all β_j . In matrix form:

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

Properties: $\text{corr}(X_j, e) = 0$ for all j ; $\text{corr}(\hat{Y}, e) = 0$.

4.3 Goodness of Fit

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}.$$

R^2 always increases as predictors are added, even random ones. The **adjusted R^2** penalises complexity:

$$R_a^2 = 1 - \frac{\text{MSE}}{\text{MST}} = 1 - \frac{\text{SSE}/(n - p - 1)}{\text{SST}/(n - 1)}.$$

Regression standard error: $s = \sqrt{\text{SSE}/(n - p - 1)}$.

4.4 Inference: t -Tests for Individual Coefficients

Each $b_j \sim N(\beta_j, s_{b_j}^2)$ approximately, so:

- 95% CI: $b_j \pm 2s_{b_j}$.
- Test statistic: $t = (b_j - \beta_{j0})/s_{b_j}$; reject if $|t| > 2$ or p-value < 0.05 .

4.5 The Overall F -Test

Tests $H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0$ simultaneously:

$$f = \frac{R^2/p}{(1 - R^2)/(n - p - 1)} = \frac{\text{SSR}/p}{\text{SSE}/(n - p - 1)} = \frac{\text{MSR}}{\text{MSE}} \sim F_{p, n-p-1}.$$

Reject H_0 if f is large ($f > 4$ is a practical rule of thumb for significance).

Inference Strategy in MLR

1. Check the overall F -test first. If not significant, stop.
2. If F is significant, examine individual t -statistics.
3. Never use multiple individual t -tests as a substitute for the F -test.

4.6 The Partial F -Test

Compares a reduced (base) model with p_{base} predictors to a full model with p_{full} predictors:

$$H_0 : \beta_{p_{\text{base}}+1} = \dots = \beta_{p_{\text{full}}} = 0.$$

$$f = \frac{(R_{\text{full}}^2 - R_{\text{base}}^2)/(p_{\text{full}} - p_{\text{base}})}{(1 - R_{\text{full}}^2)/(n - p_{\text{full}} - 1)} \sim F_{p_{\text{full}} - p_{\text{base}}, n - p_{\text{full}} - 1}.$$

Reject H_0 if f is large — the added predictors significantly improve the model.

4.7 Understanding Dependencies Among Predictors

In MLR, each partial slope b_j measures the effect of X_j *holding all other X 's fixed*. This can differ dramatically from the marginal (simple regression) slope.

Example: Price and Sales

Simple regression of Sales on own price P_1 gives a *positive* slope — surprising! Once competitor price P_2 is included, the slope of P_1 becomes negative (-97.7), correctly capturing that higher own price *holding P_2 fixed* reduces sales. The marginal slope was confounded by the positive correlation between P_1 and P_2 .

5 Multicollinearity

5.1 Definition and Consequences

Multicollinearity occurs when two or more predictors are highly correlated. In MLR the standard error of b_j depends on the variation in X_j *not* explained by the other X 's:

$$s_{b_j}^2 = \frac{s^2}{\text{SSE}_j},$$

where SSE_j is the residual sum of squares from regressing X_j on all other predictors. When predictors are highly correlated, $\text{SSE}_j \rightarrow 0$, so $s_{b_j} \rightarrow \infty$.

Consequences:

- Inflated standard errors $\Rightarrow$ wider confidence intervals.
- Slopes become unstable and sensitive to small data changes.
- Individual t -statistics can be insignificant even though the overall F is significant.

5.2 Variance Inflation Factor (VIF)

$$\text{VIF}(X_j) = \frac{1}{1 - R_j^2},$$

where R_j^2 is the R^2 from regressing X_j on all other predictors.

$$\text{se}(b_j) = \frac{s}{s_{X_j}\sqrt{n}} \cdot \sqrt{\text{VIF}(X_j)}.$$

VIF Rules of Thumb

- $\text{VIF} > 5$: moderate concern.
- $\text{VIF} > 10$: serious concern; consider removing or combining the collinear predictor.

5.3 Warning Signs of Collinearity

1. R^2 increases less than expected when adding a predictor.
2. Slopes of correlated variables change dramatically when a predictor is added/removed.
3. Strong F -statistic but weak individual t -statistics.
4. Standard errors for partial slopes larger than for marginal (simple) slopes.
5. $\text{VIF} > 5$ or 10.

5.4 Remedies

1. **Remove the redundant predictor** — retain the one with stronger theoretical justification.
2. **Combine predictors** — e.g. use the average of two highly correlated variables.

3. **Do nothing** — if estimates are still sensible and theory supports keeping both.

Key principle: statistical insignificance alone does not justify removal — subject-matter knowledge must guide variable selection.

6 Analysis of Covariance (ANCOVA)

6.1 Two-Sample Comparisons via Regression

A two-sample t -test comparing group means is equivalent to a simple regression with a **dummy variable** (indicator variable) for group membership (1 = male, 0 = female, say).

Confounding variables are correlated with both group membership and the response. Restricting analysis to subsets with matched confounders reduces bias but loses precision.

Analysis of Covariance combines dummy variables and continuous covariates to adjust for confounders while using all the data.

6.2 Adding Covariates and Interactions

To allow different intercepts and slopes per group, include:

- **Dummy variable** D (identifies group): shifts the intercept.
- **Interaction term** $D \times X$ (product of dummy and covariate): allows different slopes.

Model: $Y = \beta_0 + \beta_1 X + \beta_2 D + \beta_3 (D \times X) + \epsilon$.

Principle of Marginality:

- If the interaction is *significant*: retain it *and* both component terms.
- If the interaction is *not significant*: remove it. The resulting **parallel lines model** ($\beta_2 D$ only) is simpler to interpret.

Note: interaction terms introduce collinearity; check VIF.

6.3 Regression with Several Groups ($J > 2$)

Use $J-1$ dummy variables (one group serves as baseline). Coefficients on dummy variables represent differences in intercept vs. the baseline group. Interaction terms allow different slopes per group.

Retail Sales Example (Urban/Suburban/Rural)

Two dummy variables (Urban, Suburban; Rural = baseline). Income interacts differently with market type: urban stores have a higher intercept but lower income-sensitivity than rural stores. Population effect is constant across all locations (no interaction included).

7 Model Selection

7.1 Overfitting and the Bias–Variance Trade-off

Bias–Variance Decomposition

$$E[(y - \hat{f})^2] = \underbrace{\text{Bias}^2(\hat{f})}_{\text{systematic error}} + \underbrace{\text{Var}(\hat{f})}_{\text{estimation error}} + \underbrace{\sigma_\epsilon^2}_{\text{irreducible noise}}.$$

- **High bias (underfitting):** model too simple; misses the true relationship.
- **High variance (overfitting):** model too complex; memorises noise.
- Goal: minimise total error on *unseen* data, not training data.

Cautionary tale — Google Flu Trends (GFT): GFT selected 45 search terms from 50 million candidates to match historical CDC data, achieving 97% in-sample accuracy. In practice it overestimated flu in 100/108 weeks, and completely missed the 2009 H1N1 outbreak (a spring event outside its winter-season training pattern). Lesson: always validate on held-out data; spurious correlations are dangerous.

7.2 Three Classes of Methods

Class	Idea	When to Use
Subset Selection	Identify a subset; fit OLS on it. Hard in/out decision.	Small/moderate p ; interpretability needed.
Shrinkage (Regularisation)	Fit all p predictors but penalise large coefficients. Soft decision.	Large or collinear p .
Dimension Reduction	Project p predictors into $M < p$ combinations.	Very high p ; predictors are correlated.

7.3 Subset Selection

Best Subset Selection: fit all 2^p possible models; pick the best one per size k using SSE/R^2 , then choose among sizes using a criterion. Guaranteed to find the globally best subset, but computationally infeasible for $p \gtrsim 40$.

Forward Stepwise: start from the null model; at each step add the variable with the greatest improvement (partial F -test). Considers only $1 + p(p+1)/2$ models.

Backward Elimination: start from the full model; remove the least useful variable one at a time. Requires $n > p$.

Stepwise Regression: combines forward addition and backward elimination.

Stepwise methods are greedy — they do not guarantee finding the globally best subset, but they scale to large p where exhaustive search is impossible.

7.4 Model Selection Criteria

All four criteria estimate test error by penalising training SSE for model complexity.

Mallow's C_p :

$$C_p = \frac{\text{SSE}_p}{\text{MSE}_{\text{full}}} - (n - 2p).$$

Choose the model with p closest to C_p .

AIC (Akaike Information Criterion, Gaussian errors):

$$\text{AIC} = \frac{1}{n\hat{\sigma}^2}(\text{SSE} + 2p\hat{\sigma}^2). \quad \text{Lower is better.}$$

BIC (Bayesian Information Criterion):

$$\text{BIC} = \frac{1}{n}(\text{SSE} + \log(n)p\hat{\sigma}^2). \quad \text{Lower is better.}$$

BIC penalises complexity more heavily than AIC (uses $\log n$ instead of 2), favouring smaller models.

Adjusted R^2 :

$$R_a^2 = 1 - \frac{\text{SSE}/(n - p - 1)}{\text{SST}/(n - 1)}.$$

Cross-Validation: directly estimates test error; no assumption about model form; most reliable but computationally expensive.

8 Regularisation

8.1 Motivation

When p is large, predictors are collinear, or $p \approx n$, the OLS estimator $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ becomes unstable because $(\mathbf{X}^\top \mathbf{X})$ is nearly singular. Regularisation adds a penalty on the coefficients to reduce variance at the cost of a small increase in bias.

8.2 Ridge Regression (L_2 Penalty)

Minimise:

$$(\mathbf{Y} - \mathbf{X}\beta)^\top (\mathbf{Y} - \mathbf{X}\beta) + \lambda \beta^\top \beta.$$

Closed-form solution:

$$\hat{\beta}_{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

- $\lambda = 0$: recovers OLS.
- $\lambda \rightarrow \infty$: all $\hat{\beta}_j \rightarrow 0$.
- Coefficients are *shrunk* toward zero but **never set exactly to zero**.
- Best when all predictors are relevant; particularly useful with collinearity.

8.3 Lasso Regression (L_1 Penalty)

Minimise:

$$(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) + \lambda \sum_{j=1}^p |\beta_j|.$$

- No closed-form solution (solved by coordinate descent or LARS).
- The L_1 penalty drives small coefficients to **exactly zero** $\Rightarrow$ automatic variable selection.
- Best when the true model is sparse (few relevant predictors).

8.4 Elastic Net ($L_1 + L_2$)

Combines both penalties:

$$(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) + \lambda_1 \sum |\beta_j| + \lambda_2 \sum \beta_j^2.$$

- L_2 component groups correlated predictors together.
- L_1 component then selects among groups.
- Best when predictors are correlated *and* $p \gg n$.

8.5 Ridge vs. Lasso: Geometric Intuition

The penalty term defines a constraint region for $\boldsymbol{\beta}$:

- **Ridge:** L_2 ball (sphere). The RSS contours touch the sphere at a smooth point $\Rightarrow$ no coefficient is exactly zero.
- **Lasso:** L_1 diamond. The RSS contours typically hit a corner of the diamond where one or more $\beta_j = 0 \Rightarrow$ automatic selection.

Property	Ridge (L_2)	Lasso (L_1)
Penalty	$\lambda \sum \beta_j^2$	$\lambda \sum \beta_j $
Closed form	Yes	No
Variable selection	No	Yes
Exact zeros	No	Yes
Grouped predictors	Stable	Picks one arbitrarily
Best when	All predictors relevant	Few predictors matter

8.6 Choosing λ via Cross-Validation

1. Define a grid of λ values (e.g. 10^{-4} to 10^4).
2. For each λ , compute k -fold cross-validation MSE.
3. Choose $\lambda^* = \arg \min_{\lambda} \text{CV}(\lambda)$.
4. Re-fit the model on all data using λ^* .

One standard-error rule: choose the largest λ whose CV error is within one SE of the minimum. This gives a sparser, more interpretable model with comparable performance.

Always standardise predictors before applying Ridge or Lasso (so the penalty treats all coefficients equally).

9 Grand Summary

Topic	Key Idea	Key Formula / Tool
SLR	Line minimising squared residuals	$b_1 = r_{XY}s_Y/s_X$
Goodness of fit	Fraction of variance explained	$R^2 = SSR/SST$
SRM Assumptions	Independence, equal variance, normality	Residual plots, QQ plots
Transformations	Linearise curved relationships	Reciprocal, logarithm
Heteroscedasticity	Fan-shaped residual plot	Transform response
Autocorrelation	Dependence over time	Durbin-Watson statistic
Outliers	Leverage & influence	Fit with/without; compare
MLR	Multiple predictors; partial slopes	OLS, F -test
Multicollinearity	Correlated predictors inflate SE	VIF; remove/combine
ANCOVA	Adjust group comparisons for covariates	Dummy variables + interactions
Subset Selection	Hard in/out variable choice	Best subset, stepwise
Model criteria	Penalise complexity	C_p , AIC, BIC, Adj. R^2
Ridge	L_2 shrinkage; all variables kept	$\hat{\beta}_{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{Y}$
Lasso	L_1 shrinkage; exact zeros	Coordinate descent, λ via CV
Elastic Net	$L_1 + L_2$; correlated predictors	Two tuning parameters λ_1, λ_2

Key Takeaways

1. Check model conditions *before* interpreting output.

2. Use the F -test before individual t -tests.
3. Never compare R^2 across models with different response variables.
4. Multicollinearity inflates standard errors but does not bias coefficients.
5. Subset selection makes hard (in/out) decisions; regularisation makes soft ones.
6. Ridge keeps all predictors; Lasso performs automatic variable selection.
7. Always use cross-validation to select λ and compare methods.
8. Standardise predictors before Ridge or Lasso.
9. The bias–variance trade-off underlies all model selection: a little bias can greatly reduce variance and improve test performance.